Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam F · Form 1: Advanced reasoning

Multiple Choice Questions: Select the best answer. (1 point each)

1. Push a rightward-sliding cart at angle θ below horizontal on a surface with kinetic coefficient μ. Under what condition does increasing the push fail to increase rightward acceleration?

(A) cosθ<=μ sinθ

(B) cosθ>μ sinθ

(C) sinθ=0

(D) Always when μ=0

2. Object A, whose mass is 17 times B's, collides with B. Ignore other horizontal forces during contact. Which comparison of force magnitudes and acceleration magnitudes is correct?

(A) A exerts 17 times the force; accelerations are equal.

(B) Forces are equal; A's acceleration is 17 times B's.

(C) Different masses make both comparisons impossible.

(D) Forces are equal in magnitude; B's acceleration magnitude is 17 times A's.

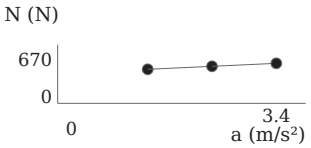
3. A 50 kg passenger stands on a scale in an elevator. Up is positive; use g=10 m/s². a (m/s²) -> N (N): (1.4, 570); (2.4, 620); (3.4, 670). If sensor gain and physical slope are both unknown, what breaks their ambiguity?

(A) Calibrate the sensor at two independently known reference values.

(B) Reverse the ordering of the data.

(C) Repeat the same unknown setting.

(D) Assume any straight graph proves exact calibration.



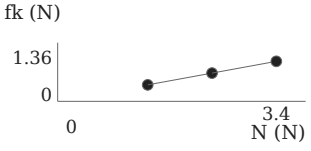
4. The same two surfaces slide against each other. Vary the normal force and measure friction magnitude. Switch to a surface with half the kinetic coefficient at the same normal loads. Plot the difference Ynew-Yold versus X. What slope should be observed?

(A) 0.052

(B) -0.2

(C) -0.116

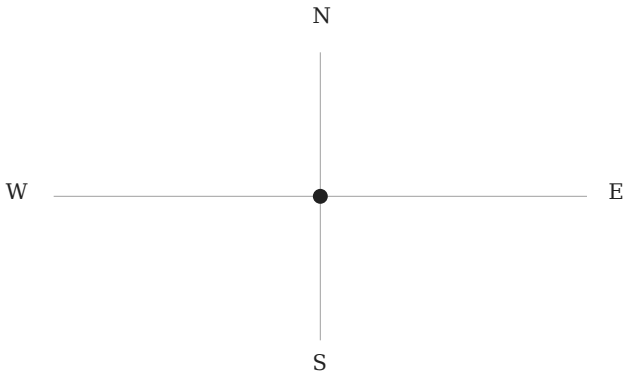
(D) -0.368



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. Two students push a cart of mass 18 kg on a frictionless horizontal floor. A exerts 380 N at 35° north of east. B exerts 510 N at 45° south of west.

a. Draw and label the actual forces acting in the plane of the floor on the dot. Draw whole vectors, not components. **(1 pt)**



b. Show net-force x and y components, then find acceleration magnitude and direction using compass words. **(3 pt)**

6. Press vertically down on a 2 kg block on a level floor. Use g=10 m/s². P (N) -> N (N): (1.4, 21.4); (2.4, 22.4); (3.4, 23.4).

a. Build the model for downward load from physics and predict both slope and intercept. **(2 pt)**

b. Double block mass, keeping the applied P itself as the horizontal coordinate. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. Hanging m1 and incline mass m2 at angle θ are connected by an ideal rope. A vertically downward F_A acts on m2. Both masses are to remain at rest. Derive minimum μ_s without assuming the friction direction. **(3 pt)**